Cover Page

DOMAIN - FINTECH

CHOSEN COMPANY - PAISSA BAZAAR

DATE - 14TH FEBRUARY 2026

CASE STUDY

DA03: Financial Risk and Performance Analysis in Consumer Lending

The FinTech industry has emerged as a dynamic and transformative force in the financial sector, integrating technology with financial services to enhance efficiency, accessibility, and customer experience. In today's era, FinTech is crucial for democratizing financial services, providing underserved populations with access to banking, credit, and investment opportunities. It fosters innovation through digital payment systems, peer-to-peer lending platforms, and blockchain technology, revolutionizing traditional banking practices.

Data scientists play a pivotal role in the growth of FinTech by leveraging advanced analytics and machine learning to improve risk assessment, fraud detection, and personalized financial services. Their expertise enables FinTech companies to analyze vast amounts of financial data, uncovering insights that drive strategic decision-making, optimize operations, and enhance customer satisfaction. By harnessing the power of data, data scientists help FinTech firms stay competitive, innovate continuously, and contribute to a more inclusive and efficient financial ecosystem.

Week I — Business Understanding & DBMS Foundations

SECTION 1 - Business Understanding Questions
Product Dissection

Question: Choose a leading platform from a domain related to the FinTech/Banking industry. Justify your selection by discussing the platform's popularity, impact, and relevance in its industry.

Answer: PaisaBazaar

Popularity

PaisaBazaar is one of India's leading online financial marketplace that allows users to apply for loans, credit cards, and insurance products from multiple banks and NBFCs — all in one place. Over the years, it has gained strong popularity among salaried

professionals, self-employed individuals, and small business owners who are looking for financial products without visiting multiple banks. Its popularity mainly comes from the convenience it offers — users can check their credit score for free, compare interest rates from different banks, and apply for loans or credit cards from a single platform. Because of its partnerships with multiple financial institutions, it has built a trusted presence in the Indian FinTech ecosystem.

Impact

PaisaBazaar has significantly changed the way people access financial products. Traditionally, customers had to visit different bank branches, meet relationship managers, and manually compare loan offers. This process was time-consuming and lacked transparency.PaisaBazaar functions as a digital intermediary that facilitates loan discovery and application submission, while final approval, verification, and disbursement are handled by partner banks. PaisaBazaar simplified this entire system by digitizing the comparison and application process. It has increased transparency in loan interest rates and credit card charges, helping users make informed financial decisions. By promoting credit score awareness and offering personalized financial recommendations, PaisaBazaar has made borrowing more structured, data-driven, and accessible.

Relevance

In today's digital era, customers expect fast approvals, minimal paperwork, and clear comparisons before choosing any financial product. PaisaBazaar is highly relevant because it addresses these expectations directly. It allows users to check their loan eligibility instantly, compare multiple financial products side-by-side, and apply digitally from home. With rising demand for personal loans, credit cards, and insurance, platforms like PaisaBazaar play an important role in financial inclusion and digital lending. It is especially relevant for young professionals who want quick and transparent financial solutions.

2. Core Features and Functionalities

Question: Research and list the core features and functionalities of the selected platform. Describe how these features contribute to the platform's success and user engagement.

ANSWER:

Core Features of PaisaBazaar

1. **Loan Comparison**: PaisaBazaar provides a comprehensive loan comparison feature that allows users to explore and compare multiple loan products offered by different banks and financial institutions based on their financial needs.

   Types of Loans Available

   - Personal loans
   - Home loans
   - Business loans
   - Car loans
   - Education loans
   - Other Need-Based Loans (medical, travel, wedding, etc.)

2. **Credit Card Comparison**: PaisaBazaar enables users to compare credit cards, check eligibility, view benefits, and digitally apply through partner banks.
3. **Credit Score Check**: PaisaBazaar allows users to check their credit score for free, understand their credit health, and access guidance on improving or resolving credit score issues.
4. **Investment Products Comparison & Information:** PaisaBazaar enables users to explore and compare investment options such as bonds, fixed deposits, and mutual funds, helping them make informed investment decisions through a single digital platform.
5. **Financial Calculators & Eligibility Tools:** PaisaBazaar's calculators help users estimate EMIs, eligibility, and investment returns in advance, enabling informed financial decisions
6. **Insurance Marketplace**: PaisaBazaar provides an insurance marketplace where users can compare and explore different insurance products such as **health insurance, life insurance, and motor insurance** offered by multiple insurers. This feature helps users understand policy coverage, premiums, and benefits before selecting a suitable insurance plan.
7. **Instant Loan Eligibility Check**: PaisaBazaar offers an instant loan eligibility check that allows users to quickly assess whether they qualify for a loan based on factors such as income, credit score, and financial profile. This reduces the chances of rejection and helps users apply for suitable loan offers, including

pre-approved options.

8. **Digital Documentation**: PaisaBazaar enables digital documentation by allowing users to complete **online KYC** and upload required documents electronically. This feature minimizes paperwork, speeds up the application process, and provides a convenient, paperless experience.
9. **Personalized Recommendations**: PaisaBazaar provides personalized product recommendations based on user data such as income, credit score, and financial profile. This helps users discover the most relevant loan, credit card, insurance, or investment options tailored to their needs.

Contribution to Success and User Engagement

Convenience:
Users can compare multiple banks on one platform.

Transparency:
Clear interest rate comparison builds trust.

Financial Awareness:
Free credit score service increases engagement.

Personalization:
AI-based recommendations improve conversion rates.

Time Saving:
Online application avoids physical branch visits.

Trust & Security:
Secure document upload and verified bank partnerships build credibility.

These features make PaisaBazaar a strong FinTech marketplace platform.

3. Real World Problems

Question: Identify the real-world problems that the platform aims to solve. Discuss how the platform addresses these problems through its features and functionalities.

ANSWER: Real-World Problems Addressed by Paisa Bazaar

Problem 1: Lack of Transparency in Loan Offers

Many customers do not clearly understand interest rates, hidden charges, or processing fees when applying for loans. This lack of transparency often leads to poor financial decisions.

Solution:
PaisaBazaar solves this by showing side-by-side comparisons of loan products with detailed information about interest rates, fees, and tenure. This allows users to make informed decisions.

Problem 2: Low Awareness About Credit Score

Many people are unaware of their credit score and how it affects loan approvals. As a result, their applications often get rejected.

Solution:
PaisaBazaar provides free credit score checks and personalized improvement tips. This increases financial literacy and improves approval chances.

Problem 3: Time-Consuming Traditional Loan Applications

Visiting banks, filling physical forms, and submitting documents manually is time-consuming and inconvenient.

Solution:
The platform allows online application and digital document upload, reducing the need for physical visits and saving time.

Problem 4: Difficulty in Comparing Financial Products

Customers often struggle to compare multiple credit cards or insurance plans properly.

Solution:
PaisaBazaar organizes financial products in a structured comparison format, making evaluation simple and easy to understand.

Problem 5: Limited Access to Financial Services in Smaller Cities

People in smaller towns may not have access to multiple bank branches or financial advisors.

Solution:
Since PaisaBazaar operates digitally, anyone with a smartphone and internet access can explore and apply for financial products from anywhere.

How PaisaBazaar Addresses These Problems

- Provides centralized comparison platform
- Uses AI to check eligibility instantly
- Offers digital KYC process
- Provides credit score awareness tools
- Connects users with multiple partner banks

These features and functionalities collectively address significant real-world problems, contributing to Paissa Bazar's success and high user satisfaction.

SECTION 2 - Schema Design

4. Schema Design

Question: Based on the features and functionalities you have identified, design a schema that reflects the platform's data structure. Define the key entities, attributes, and relationships that underpin these features.

Answers: Schema Design for Paissa Bazzar

The schema design for Passia Bazaar should reflect its core features and functionalities, ensuring efficient data management and enabling the platform to deliver a seamless user experience. Below is an overview of the key entities, attributes, and relationships.

Key Entities and Attributes

1. User Table

Attribute	Type	Description
userID	Primary Key (UUID)	Unique user identifier
fullName	String	Name of the user
email	String	Registered email
phoneNumber	String	Contact number
passwordHash	String	Encrypted password
creditScore	Integer	User's credit score
monthlyIncome	Decimal	Monthly income
employmentType	String	Salaried / Self-employed
registrationDate	Date	Account creation date

2. Bank Table

Attribute	Type	Description
bankID	Primary Key	Unique bank identifier
bankName	String	Name of financial institution
ProviderType	string	Specifies whether the provider is a bank, NBFC, or insurance company

3.Loan Product Table

Attribute	Type	Description
loanID	PK	Unique loan product ID
bankID	FK → Bank	Loan provider
loanType	String	Personal / Home / Business
interestRate	Decimal	Loan interest rate
maxAmount	Decimal	Maximum loan amount
tenure	Integer	Loan duration (months)
processingFee	Decimal	Processing charges
eligibilityCriteria	String	Loan eligibility

4. Loan Application Table

Attribute	Type	Description
applicationID	Primary Key	Unique application ID
userID	Foreign Key	Linked to User
loanID	Foreign Key	Selected loan product
appliedAmount	Decimal	Requested loan amount
applicationDate	Date	Date of application
applicationStatus	String	Pending / Approved / Rejected

5. Credit Card Table

Attribute	Type	Description
cardID	Primary Key	Unique card ID
bankID	Foreign Key	Issuing bank
cardName	String	Name of card
annualFee	Decimal	Yearly fee
rewardType	String	Cashback / Travel
interestRate	Decimal	Interest on dues

6. Credit Card Application Table

Attribute	Type	Description
cardAppID	Primary Key	Unique application ID
userID	Foreign Key	Linked to User
cardID	Foreign Key	Selected credit card
applicationDate	Date	Application date
status	String	Approved / Pending / Rejected

7. Insurance Product Table

Attribute	Type	Description
insuranceID	Primary Key	Unique insurance ID
bankID	Foreign Key	Linked to Bank

insuranceType	String	Health / Life / Motor
premiumAmount	Decimal	Insurance premium
coverageAmount	Decimal	Coverage value

Insurance Application Table

Attribute	Type	Description
insuranceAppID	Primary Key	Unique insurance application ID
userID	Foreign Key	Linked to User
insuranceID	Foreign Key	Selected insurance product
coverageSelected	Decimal	Coverage amount chosen by the user
applicationDate	Date	Date of insurance application
applicationStatus	String	Pending / Approved / Rejected

Relationship Table

From Table	FromAttribute	To Table	To Attribute	RelationsType
User	userID	LoanApplication	userID	One-to-Many
User	userID	CreditCardApplication	userID	One-to-Many
User	userID	InsuranceApplication	userID	One-to-Many
Bank	bankID	LoanProduct	bankID	One-to-Many
LoanProduct	loanID	LoanApplication	loanID	One-to-Many
Bank	bankID	CreditCard	bankID	One-to-Many
CreditCard	cardID	CreditCardApplication	cardID	One-to-Many
Bank	bankID	InsuranceProduct	bankID	One-to-Many
InsuranceProduct	insuranceID	InsuranceApplication	insuranceID	One-to-Many

This schema design captures the essential data interactions and relationships within Paissa Bazzar , supporting its features and functionalities efficiently. It ensures scalability, performance, and a personalized user experience, contributing to the platform's overall success.

5. ER Diagram Creation

Question: Utilise tools like the Miro platform or similar applications to create an illustrative Entity-Relationship (ER) diagram. This diagram should vividly depict the entities, attributes, and relationships present within your schema design.

Revenue and Profit Growth Strategies

Question: After completing the product dissection and schema design steps for the chosen platform, conduct a comprehensive case study on the above chosen industry. Your goal is to identify and propose strategies to increase the profit of the industry by at least 25%.

Create a detailed report summarising your findings and proposals. Include data-driven justifications for each proposed strategy and present your case study using visual aids such as charts, graphs, and diagrams to illustrate your points. Outline the steps, resources, and timeline required to achieve the desired revenue and profit growth.

I. Analysing PaisaBazaar's Current Status

To increase PaisaBazaar's profit by 25%, we must first analyse its internal structure, revenue model, cost drivers, and customer behaviour. Since PaisaBazaar operates as a financial marketplace (not a lender itself), its profitability depends mainly on commissions, partnerships, and operational efficiency.

Category	Details
Current Financial Data	**Collect Revenue Data:** Gather data from all revenue streams including loan commissions, credit card commissions, insurance commissions, and partner bank payouts. Since Paisabazaar operates as a financial marketplace, most revenue comes from financial institutions for customer acquisition. **Collect Expense Data:** Include technology infrastructure costs, marketing and advertising spend, employee salaries, customer support operations, and compliance/legal expenses. **Analyze Current Profit:** Calculate net profit by subtracting operational and marketing expenses from total commission-based revenue.
Profit Trends	Analyze profit trends over the last 2–3 years to understand growth patterns. Identify whether revenue growth is driven by higher loan volumes, increased partnerships, or improved customer acquisition

	efficiency. Detect seasonal demand spikes (e.g., festive loan demand).
Sources of Revenue	**Loan Commissions:** Revenue earned when users successfully apply for personal loans, home loans, or business loans through the platform. **Credit Card Commissions:** Commission from banks for each approved credit card application. **Insurance Partnerships:** Earnings from selling health, life, or motor insurance products. **Lead Generation Services:** Selling verified financial leads to partner institutions. **Advertising & Featured Listings:** Banks and NBFCs pay for premium placement and visibility on the platform.
Revenue Contribution Analysis	Segment revenue based on product type (loans vs credit cards vs insurance). Identify which product category contributes the highest margin and which segment has growth potential. This helps focus marketing and product development efforts.
Sources of Expenses	**Marketing & Customer Acquisition:** Paid ads, digital campaigns, SEO, influencer partnerships. **Technology Infrastructure:** Platform maintenance, data analytics systems, cybersecurity, cloud services .**Employee Salaries:** Sales teams, data analysts, engineers, customer support staff. **Compliance & Legal Costs:** Regulatory compliance, data protection, and financial regulations. **Operational Costs:** Office infrastructure, vendor services, CRM tools.
Customer Acquisition & Retention	**Acquisition Channels:** Digital ads (Google, Facebook), affiliate marketing, referral programs, SEO traffic. **Customer Behavior Analysis:** Study loan comparison patterns, drop-off rates during applications, and product preferences. **Retention Rate:** Measure repeat usage for different financial products. **Churn Analysis:** Identify reasons why customers don’t complete applications (complex forms, slow processing, lack of trust).

II. Focus Areas for Increasing PaisaBazaar's Profit by 25%

To increase Paisa Bazaar’s profit by 25%, the company must strategically focus on key performance-driving areas such as operational efficiency, revenue growth, customer retention and engagement, and brand and market expansion. As a digital financial marketplace operating on a commission-based model, profitability depends heavily on optimizing internal processes, increasing high-margin product conversions, improving customer lifetime value, and expanding into high-growth markets. By implementing targeted, data-driven initiatives across these areas, Paisa Bazaar can enhance operational effectiveness, improve approval rates, strengthen customer trust, and expand its market presence in the competitive fintech industry.

Category	Focus Area	Measures
Operational Efficiency (~8%)	Process Automation (~3%)	Implement AI-based credit profiling and automated eligibility matching to reduce manual review time. Automate document verification using OCR and API-based bank integrations. Reduce turnaround time to improve customer satisfaction and reduce operational costs.
	Cost Optimization (~2%)	Reduce high dependency on paid advertisements by improving SEO and content marketing. Optimize marketing budgets toward high-converting digital channels. Conduct regular audits of technology tools and renegotiate vendor contracts.
	Compliance & Risk Management (~3%)	Strengthen fraud detection using AI-based risk assessment tools. Ensure strict RBI compliance and data security measures to avoid penalties. Reduce bad-quality leads by improving verification filters.
Revenue Growth (~9%)	High Commission Products (~4%)	Promote high-commission financial products such as premium credit cards, personal loans, and business loans. Prioritize partnerships with banks offering higher payout rates. Focus marketing campaigns on high-margin financial products.

	Cross-Selling & Upselling (~3%)	Offer bundled financial products such as loan + insurance or credit card + credit score subscription. Use customer financial history to suggest relevant additional services. Increase average revenue per customer.
	New Revenue Streams (~2%)	Introduce premium subscription-based financial advisory services. Launch paid credit improvement programs and financial planning tools for users seeking detailed guidance.
Customer Retention & Engagement (~5%)	Personalization (~2%)	Use AI-driven recommendation systems to show customized financial products based on income, credit score, and spending patterns. Improve approval success rate by matching users accurately.
	Loyalty & Referral Programs (~1.5%)	Introduce referral rewards for users who bring new customers. Provide cashback or reward points for successful loan disbursals or credit card approvals. Encourage repeat usage.
	Customer Support Enhancement (~1.5%)	Improve chatbot efficiency for instant financial queries. Provide dedicated relationship managers for high-value customers. Reduce response time to improve trust and retention.
Market Expansion & Branding (~3%)	Tier 2 & Tier 3 Market Penetration (~2%)	Expand digital financial awareness campaigns in smaller cities where credit demand is rising. Partner with local financial advisors to increase trust and adoption.
	Trust Building & Branding (~1%)	Run campaigns highlighting RBI compliance, secure data handling, and transparent comparison services. Build brand positioning as a trusted financial marketplace.

By concentrating on operational efficiency, revenue growth, customer retention, and market expansion, Paisa Bazaar can systematically achieve a 25% increase in profitability. Each focus area should be guided by advanced analytics, automation, and customer behavior insights to ensure that strategies remain aligned with financial trends and user expectations. Implementing these initiatives will not only improve revenue generation and cost optimization but will also strengthen Paisa Bazaar’s competitive positioning, brand credibility, and long-term sustainability in India’s rapidly growing digital financial services market.

III. Defining Strategies

Category	Details
Optimize Expenses	**Marketing Cost Optimization:** Shift budget toward high-conversion digital channels and reduce dependency on low-performing paid campaigns. Strengthen SEO and organic acquisition to reduce Cost per Lead (CPL). **Process Automation:** Automate document verification, lead routing, and customer support using AI to reduce manual workload and operational costs. **Data-Driven Cost Monitoring:** Implement real-time dashboards to track CAC (Customer Acquisition Cost) and operational expenses for continuous optimization.
Enhance Revenue Streams	**High-Commission Product Focus:** Prioritize promotion of high-margin financial products such as premium credit cards, unsecured personal loans, and insurance bundles.Upselling & **Cross-Selling:** Use behavioral analytics to recommend complementary financial products (e.g., loan + insurance). **Dynamic Partner Strategy:** Strengthen relationships with banks/NBFCs offering higher payout structures. **New Revenue Streams:** Introduce subscription-based credit score monitoring or premium financial advisory services.
Improve Customer Satisfaction & Retention	**AI-Powered Personalization:** Provide instant eligibility results and personalized product recommendations based on credit profile. **Loyalty & Referral Programs:** Offer rewards for completed applications and successful referrals to increase repeat usage.

	Transparent Communication: Clearly explain interest rates, processing fees, and approval criteria to build trust and reduce drop-offs.
	Customer Support Optimization: Use chatbots + trained advisors for faster resolution.
Brand & Market Expansion Strategy	**Tier 2 & Tier 3 Market Penetration:** Expand into underpenetrated cities with growing digital adoption.
	Localized Campaigns: Run regional-language marketing campaigns to build trust among new users.
	Trust-Based Branding: Emphasize data security, RBI compliance, and transparent comparison models in marketing communications.
Technology & Innovation Strategy	**Predictive Credit Matching:** Use machine learning models to match users with higher approval probability lenders to increase conversion rates.
	Fraud Detection & Risk Control: Implement stronger cybersecurity and fraud analytics systems.
	Mobile Experience Optimization: Improve app speed and reduce application steps to increase completion rate.
	Advanced Analytics: Track approval rates, drop-off stages, and lifetime value (LTV) to continuously refine strategy.

By adopting an inside-out approach, Paisa Bazaar can systematically improve its profitability. The strategy begins with optimizing operational and marketing costs, followed by enhancing customer satisfaction through personalization and transparent financial comparisons. It then focuses on increasing revenue through high-commission partnerships, cross-selling, and premium financial services. Using data-driven insights such as conversion rates and customer acquisition cost ensures smarter decision-making. This structured approach will help Paisa Bazaar achieve its targeted 25% profit growth sustainably while strengthening its market position.

SECTION 3 - Guesstimate Questions(Interview-Focused)

1. What will be the percentage increase in global FinTech investments over the next five years?

Step	Details	Calculation
Current Year Investment	Present global FinTech investment	₹100 crore
Growth Rate	Expected annual growth	20%
Growth Factor	Annual growth multiplier	1 + 0.2 = 1.2
Investment Period	Duration of growth	5 years
Final Investment Value	Investment after 5 years	₹100 × $(1.2)^5$ = ₹249 crore
Increase in Investment	Additional investment	₹249 − ₹100 = ₹149 crore
Percentage Increase	Growth over 5 years	**149%**

Near about **149%** growth is expected over the next 5 years if we are considering current growth as 100cr.

2. How many people will adopt digital banking services in developing countries over the next decade?

   This question requires an estimation of the number of new users of digital banking solutions in regions where traditional banking infrastructure is less prevalent.

Step-by-Step Population Breakdown

Step	Category	Assumption	Calculation	People
1	Total Population (India)	Approximate current population	—	1.4 billion
2	Total Population (India)	Converted to millions	1.4 × 1,000	1,400 million
3	Adult Population (18+)	~65% are adults	1,400 × 65%	910 million
4	Internet / Smartphone Users	~70% have access	910 × 70%	637 million
5	Currently NOT Using Digital Banking	~55% still non-digital	637 × 55%	350 million (approx.)

Near about **350** million potential **new digital banking users** over time

3. What percentage of small and medium-sized enterprises (SMEs) will use FinTech solutions for their financial needs by 2025?

This question involves predicting the adoption rate of FinTech services among SMEs, including payments, lending, and financial management tools.

Step	Category	Assumption	Calculation	People
1	**Total Population (India)**	Current population	—	**1.4 billion**
2	**Total Population (in millions)**	Unit conversion	1.4 × 1,000	**1,400 million**
3	**Working-age Population**	60%	1,400 × 60%	**840 million**
4	**Business Owners / Entrepreneurs**	10% of working population	840 × 10%	**84 million**
5	**SMEs (formal + informal)**	70% of businesses	84 × 70%	**58.8 million**

6	**SMEs with Digital Access**	Internet / smartphone access ≈ 80%	58.8 × 80%	**47.0 million**
7	**SMEs Using FinTech by 2025**	Expected adoption ≈ 65%	47.0 × 65%	**30.6 million**

Adoption Rate (%) = (SMEs using FinTech ÷ SMEs with digital access) × 100

Adoption Rate (%) = (30.6 ÷ 47.0) × 100 ≈ 65.1%

Approximately 65% of SMEs are expected to use FinTech solutions for their financial needs by 2025.

4. What will be the average transaction value of mobile payments in the next three years?

This question requires estimating the average amount of money transacted through mobile payment platforms, considering the growth of mobile commerce and digital

Step	Description	Calculation	Value
1	Current Avg Transaction Value	Assumed present value	₹1,500
2	Annual Growth Rate	Inflation + higher digital usage	8%

3	Time Period	Future estimation	3 years
4	Growth Factor	Convert % to factor	1 + 0.08 = **1.08**
5	Compound Growth (3 Years)	$1500 \times (1.08)^3$	**₹1,890**

The average transaction value of mobile payments is expected to be around ₹1,900 in the next three years.

5. How much will blockchain technology reduce the costs of cross-border transactions in the next five years?

This question involves estimating the cost savings achieved through the adoption of blockchain technology for international money transfers, factoring in current fees and the efficiency improvements brought by blockchain.

Step	**Description**	**Calculation**	**Result**
1	Current transaction fee	Given assumption	**6%**

2	Target fee reduction over 5 years	Assumed improvement due to blockchain	**3%**
3	Transaction fee after 5 years	6% − 3%	**3%**
4	Total cost reduction	Initial − Final	**3%**
5	Percentage reduction	(3 ÷ 6) × 100	**50%**

Blockchain technology can reduce cross-border transaction costs from around 6% today to about 3% over the next five years, resulting in an approximate 50% reduction in costs.

SECTION 4 - Cohort Analysis & A/B Testing Scenario Based Questions

Scenario 1:A fintech company offers a mobile payment app where users can link their bank accounts and make payments. The company wants to analyse the retention of users who signed up for the app in the past six months to understand how often they continue using it after the initial download.

Question 1:
How would you perform a cohort analysis to calculate the monthly retention rate for users who signed up in different months?

- Hint: Group users by the month they signed up, and track their activity (i.e., whether they made any transactions) in subsequent months.

ANSWER:

To analyze user retention for Paisa Bazaar's mobile payment and financial services app, a cohort analysis approach can be used. The objective is to measure how many users

who signed up in a particular month continue to perform financial transactions in the following months. This helps evaluate user engagement, retention patterns, and churn trends.

Step 1: Define the Cohorts

Users will be grouped based on the **month of sign-up**. Each cohort represents users who registered in the same calendar month.

For example:

Cohort (Sign-Up Month)	Number of New Users
January 2025	1,000
February 2025	1,200
March 2025	900
April 2025	1,100
May 2025	1,300
June 2025	1,000

Each group will be tracked separately over the following months.

Step 2: Define Retention Metric

For Paisa Bazaar, **Monthly Retention Rate** can be defined as:

> The percentage of users from a cohort who perform at least one financial transaction (loan application, credit card application, EMI payment, or bill payment) in a given month after sign-up.

Step 3: Track Monthly Activity

For each cohort, track whether users performed any transaction in subsequent months.

Example Retention Table:

Cohort	Month 0 (Signup)	Month 1	Month 2	Month 3	Month 4	Month 5
Jan 2025 (1,000 users)	100%	65%	52%	40%	32%	25%
Feb 2025 (1,200 users)	100%	68%	55%	43%	35%	28%
Mar 2025 (900 users)	100%	70%	58%	47%	38%	30%

Apr 2025 (1,100 users)	100%	66%	54%	42%	34%	—
May 2025 (1,300 users)	100%	69%	56%	—	—	—
Jun 2025 (1,000 users)	100%	72%	—	—	—	—

Step 4: Analyse Patterns

From this cohort table, Paisa Bazaar can observe:

- Retention typically drops after Month 1 (natural early churn).
- March and June cohorts show slightly higher Month 1 retention → possibly due to better marketing campaigns or improved onboarding.
- Retention declines gradually over months, indicating decreasing repeat engagement.

This pattern helps identify:

- Churn rate
- Engagement lifecycle
- Effectiveness of onboarding experience

- Seasonal trends in financial product demand

Step 5: Derive Actionable Insights

Based on cohort retention analysis, Paisa Bazaar can:

1. Improve onboarding experience to reduce Month 1 drop-off.
2. Introduce reminder notifications and personalized offers in Month 2 and Month 3.
3. Launch loyalty rewards for repeat applications.
4. Improve matching algorithms to increase approval success rate.
5. Retarget inactive users through email and app campaigns.

Business Importance for Paisa Bazaar

Cohort analysis helps Paisa Bazaar:

- Measure customer lifetime value (LTV)
- Identify churn periods
- Evaluate effectiveness of marketing campaigns
- Improve product engagement strategy
- Increase long-term profitability

By grouping users based on sign-up month and tracking their transaction activity over time, Paisa Bazaar can accurately calculate monthly retention rates and make data-driven strategic decisions to improve user engagement and reduce churn.

Question 2:

If you find that retention drops significantly after the first month, what are some possible reasons for this behaviour, and how can the company address this drop-off?

- Hint: Consider user engagement strategies such as onboarding improvements, notifications for unused accounts, or offering discounts for making a second transaction.

ANSWER:

If the cohort analysis shows that user retention drops significantly after the first month, it indicates that many users download the app, complete an initial transaction (such as checking loan eligibility or applying for a credit card), but do not return for further activity. This behaviour is common in fintech platforms where users may join with a single financial objective.

Possible Reasons for Retention Drop

1. One-Time Purpose Usage

Many users may download Paisa Bazaar to compare loans or apply for a specific financial product. Once their need is fulfilled, they may not see an immediate reason to return.

2. Poor Onboarding Experience

If the onboarding process is lengthy, confusing, or requires excessive documentation, users may disengage after their first interaction.

3. Low Approval or Conversion Rate

If users face rejection in loan or credit card applications, they may lose trust and stop using the platform.

4. Lack of Engagement After First Transaction

If there are no follow-up notifications, reminders, or personalized recommendations, users may forget about the app.

5. Trust and Transparency Issues

In fintech, unclear fees, complex financial terms, or hidden conditions may discourage repeat usage.

How Paisa Bazaar Can Address the Drop-Off

1. Improve Onboarding Experience

- Simplify sign-up and KYC processes.
- Provide instant eligibility checks.
- Offer guided assistance during the first transaction.

2. Strengthen Engagement Strategies

- Send personalized notifications for new financial offers.
- Remind users about credit score updates or better interest rates.
- Provide financial tips through in-app messages.

3. Incentivize Second Transaction

- Offer cashback or reward points for completing a second transaction.
- Provide limited-time discounts on processing fees.
- Launch loyalty benefits for repeat applications.

4. Enhance Personalization

- Use AI-based matching to show only highly relevant financial products.
- Recommend complementary products (e.g., insurance after loan approval).

5 .Build Trust and Transparency

- Clearly display interest rates, charges, and eligibility criteria.
- Highlight RBI compliance and data security practices.

Strategic Conclusion

A sharp drop after Month 1 suggests early-stage churn. By improving onboarding, increasing engagement, incentivizing repeat usage, and strengthening trust, Paisa Bazaar can improve monthly retention rates. Enhancing retention directly increases customer lifetime value (LTV), reduces customer acquisition cost (CAC) pressure, and supports sustainable profit growth.

Scenario 2:

The fintech company is testing two different loan approval notification designs. Version A is a simple approval message, while Version B includes additional loan details (e.g., repayment options, interest rate, and payment reminders). They want to see which design leads to more loan acceptance.

Question 1:

How would you structure an A/B test to measure the impact of these notification designs on loan acceptance rates?

- Hint: Track loan acceptance rates (percentage of users who accept the loan offer), average loan amount, and repayment behaviours after loan acceptance.

ANSWER:

Scenario Overview

Paisa Bazaar wants to test two different loan approval notification designs in its mobile app to determine which version leads to higher loan acceptance rates.

- **Version A (Control):** Simple loan approval message – “Your loan has been approved.”

- **Version B (Variant):** Detailed loan approval message including:

    - Approved loan amount

    - Interest rate

    - EMI breakdown

    - Repayment schedule

- Payment reminders

The objective is to evaluate which version improves **loan acceptance, engagement, and repayment behaviour**.

Step 1: Hypothesis Formation

Hypothesis:

Providing detailed loan information (Version B) will increase loan acceptance rates and improve repayment behaviour compared to a simple approval message (Version A), because users feel more informed and confident about the financial commitment.

Step 2: Create Variants

Version	Description
Version A (Control)	Simple approval message with minimal details
Version B (Variant)	Detailed approval message with repayment options, EMI breakdown, and reminders

Step 3: Random Assignment

- Randomly divide eligible approved users into two equal groups.
- 50% of users receive Version A.
- 50% of users receive Version B.
- Ensure isolation from external influences (same time period, same loan type, similar risk profile).

This prevents bias and ensures accurate comparison.

Step 4: Define Experiment Duration

- Run the test for 30–45 days.
- Ensure statistically significant sample size.
- Avoid major marketing or UI changes during the experiment period.

Step 5: Define Key Metrics (KPIs)

To measure impact, track:

1. Primary Metric – Loan Acceptance Rate

2 .Secondary Metrics

- Average Loan Amount Accepted
- Time Taken to Accept Loan
- Repayment Behaviour (EMI on-time payment rate)
- Early Drop-off Rate

Step 6: Data Collection (Example)

Variant	Users Notified	Loans Accepted	Acceptance Rate	Avg Loan Amount	On-Time EMI Rate
Version A	10,000	4,000	40%	₹2,00,000	82%
Version B	10,000	5,200	52%	₹2,20,000	88%

Step 7: Analysis

From the example:

- Version B increases acceptance rate from 40% to 52%
  → 30% relative improvement
- Higher average loan amount
- Better repayment behaviour

This indicates that providing transparency and detailed financial information increases user trust and confidence.

Step 8: Conclusion

If Version B consistently shows statistically significant improvement across acceptance rate and repayment performance:

- Implement Version B permanently.
- Standardize detailed notifications for all loan approvals.
- Further optimize repayment reminders for even better retention.

Business Impact for Paisa Bazaar

Implementing the better-performing notification design can:

- Increase conversion rate (CRO improvement)
- Increase commission revenue (since commission is earned on successful disbursals)
- Improve repayment quality (lower default risk for partner banks)
- Strengthen customer trust and transparency
- Improve long-term customer lifetime value (LTV)

Final Strategic Insight

A/B testing in this scenario enables Paisa Bazaar to make data-driven decisions rather than relying on assumptions. By continuously testing and optimizing user communication, the company can improve loan acceptance rates, enhance user confidence, and drive sustainable revenue growth in a competitive fintech environment.

Question 2:

Suppose Version B (detailed notification) results in a 10% increase in loan acceptance rates but requires additional resources to implement. How would you evaluate whether the increase in acceptance rates justifies the added complexity?

- Hint: Consider the potential revenue increase from higher loan acceptance rates and whether the costs of implementing Version B are outweighed by the financial benefits. You might also suggest simplifying some of the details to reduce complexity.

ANSWER:

Scenario

From the A/B test:

- Version B (detailed notification) increases loan acceptance rate by **10%**
- However, it requires:
    - Additional UI/UX development
    - Backend integration for EMI breakdown & repayment schedule
    - Ongoing maintenance
    - Possible notification system upgrades

Step 1: Quantify the Revenue Impact

Paisa Bazaar earns revenue primarily through:

- Commission from partner banks/NBFCs on successful loan disbursal
- Processing fees
- Cross-selling financial products

Example Calculation

Assume:

- 100,000 users receive loan approval notifications monthly
- Version A acceptance rate = 40%
  → 40,000 loans accepted
- Version B acceptance rate = 44% (10% relative increase from 40%)

Additional accepted loans:
= 4,000 extra loans per month

If:

- Average loan size = ₹2,00,000
- Commission rate = 1%

Revenue per loan:
₹2,00,000 × 1% = ₹2,000

Additional monthly revenue:
4,000 × ₹2,000 = ₹80,00,000 (₹80 lakh per month)

Annual additional revenue:
₹80 lakh × 12 = ₹9.6 crore per year

Step 2: Estimate the Cost of Implementation

Now evaluate total cost:

One-Time Costs

- UI/UX redesign
- Backend development
- Testing & QA
- Analytics integration

Example: ₹50–70 lakh

Recurring Costs

- Server costs
- Maintenance

- Notification infrastructure
- Customer support

Example: ₹10–15 lakh annually

Step 3: Compare Revenue vs Cost

If:

- Annual additional revenue = ₹9.6 crore
- Total cost (Year 1) ≈ ₹1 crore

Net benefit (Year 1) ≈ ₹8.6 crore

ROI is significantly positive.

Thus, the 10% improvement clearly justifies implementation.

Step 4: Evaluate Beyond Revenue

In fintech, we must also consider:

1 Repayment Behaviour

If Version B also improves:

- EMI on-time payments
- Reduces default rates

Then:

- Partner banks gain trust
- Paisa Bazaar strengthens relationships
- Long-term revenue increases

2 Customer Trust & Transparency

Providing loan details:

- Reduces confusion
- Reduces customer complaints
- Improves brand credibility

This has strong long-term strategic value.

Step 5: Sensitivity & Risk Analysis

However, before full rollout:

- Check statistical significance
- Ensure the 10% lift is sustainable
- Run the test across different user segments

- Check if higher acceptance leads to higher default risk

Step 6: Optimization Strategy (If Cost Is Too High)

If implementation complexity is too expensive, Paisa Bazaar can:

- Simplify the detailed message (show key highlights only)
- Use expandable sections instead of full UI redesign
- Use dynamic templates instead of backend-heavy logic
- Phase rollout (start with high-value loan segments)

This reduces development complexity while maintaining impact.

Final Decision Framework

To justify Version B, Paisa Bazaar should calculate:

Net Business Impact=Additional Revenue−Implementation Cost

If:

- Net Impact > 0
- ROI is strong
- Repayment behaviour improves

- Brand trust increases

Then Version B should be implemented.

Final Conclusion

Although Version B requires additional resources, the 10% increase in loan acceptance can generate substantial incremental revenue for Paisa Bazaar. If the financial gains significantly exceed development and maintenance costs — and repayment quality remains stable — the added complexity is justified.

A data-driven ROI evaluation ensures that the decision is financially sound, strategically aligned, and sustainable for long-term business growth.

SECTION 5- DBMS & Data Architecture

ACID Properties:

1. Fund transfer crashes midway

Q: A user initiates a fund transfer from Account A to Account B, but the app crashes midway. How would you ensure that no partial transaction corrupts the account balances?

A:

Atomicity: The fund transfer is treated as a single transaction. Either both debit from Account A and credit to Account B happen together, or none of them happen. If the app crashes, the database rolls back the transaction.

Durability: Once the transfer is successfully completed, the transaction is permanently saved in the database even if the system crashes later.

This ensures PaisaBazaar never stores half-completed financial transactions.

2. Two traders place orders at the same time

Q: How would you ensure that each transaction executes reliably without interference?

A:

Isolation: Each transaction runs independently so one user's transaction does not affect another's.

Consistency: The database ensures that rules (such as account balance and stock quantity) remain valid after every transaction.
In PaisaBazaar, concurrent loan or payment requests are handled safely using transaction isolation levels to avoid data conflicts.

Indexing:

1. Real-time dashboards for high-net-worth clients

Q: How can indexing help here?

A:
Indexes can be created on frequently queried columns such as userID, transactionDate, and accountID.
This allows PaisaBazaar to fetch balances and transaction history quickly for dashboards instead of scanning the entire table.

2. Transaction data queried by ID and date

Q: What indexing strategies would you use?

A:

Use B-Tree index on transactionID for fast lookups.

Use a composite index on (transactionDate, userID) for queries filtered by date and user.

This improves performance while keeping update costs manageable.

Partitioning:

1. Billions of transaction records

Q: How would you design it for efficient querying and archiving?

A:
Use time-based (range) partitioning such as monthly or yearly partitions.
Each partition stores transactions for a specific time period, making queries faster and archiving old data easier.

2. Data residency laws

Q: How can partitioning help?

A:
Use geo-partitioning based on region (India, EU, US).
Customer data is stored in regional partitions to comply with legal and regulatory requirements.

Sharding:

1. Peak hour microtransactions

Q: How would you use sharding to scale horizontally?

A:
PaisaBazaar can apply horizontal sharding by customerID.
User data is distributed across multiple database servers, reducing load on a single server and improving performance during peak hours.

2. Global transaction analytics

Q: What problems might you face?

A:
Sharding by customerID makes global analytics difficult because data is spread across shards.
Cross-shard joins and aggregations become slow and complex.
To solve this, data is copied into a centralized data warehouse for analytics.

Normalization and Denormalization:

1. Customer profiles, accounts, and KYC documents

Q: How would normalization help?

A:
Normalization separates data into multiple tables such as User, Account, and KYC_Documents. This avoids duplicate data, improves data integrity, and ensures consistent customer information across PaisaBazaar systems.

2. Monthly financial reporting

Q: Normalized or denormalized schema?

A:
A denormalized schema is preferred for reporting because it reduces the number of joins and improves read performance.
Summary tables combining user, loan, and transaction data can be used for faster reports.

Data Lakes vs. Data Warehouses:

1. Where would you store different data types?

Q: You receive unstructured logs, real-time clickstream data, and structured account data. Where would you store each, and why?

A:

Data Lake: Unstructured logs and real-time clickstream data (raw format).

Data Warehouse: Structured account, loan, and transaction data for analytics and reporting.

2. Fraud detection and risk analysis

Q: How does a data warehouse help?

A:
A data warehouse supports fraud detection using historical transaction data and OLAP queries. It enables pattern analysis, trend detection, and risk modeling using aggregated data over time.

Data Ingestion and Pipelines:

1. Real-time payments data

Q: How would you ingest this data?

A:
Use streaming tools such as Apache Kafka or AWS Kinesis to ingest payment data from mobile apps in real time.
This allows continuous monitoring for fraud detection and alerts.

2. Daily credit score data

Q: What ingestion approach would you use?

A:
Use batch ingestion with scheduled ETL jobs (daily).
Data from third-party credit bureaus is imported once per day and stored in PaisaBazaar's database.

ETL (Extract, Transform, Load):

1. Consolidating investment portfolios

Q: How would you design the ETL pipeline?

A:

Extract: Collect data from multiple banking systems.

Transform: Clean data, remove duplicates, map fields to a common schema.

Load: Store transformed data in a central analytics platform for reporting and insights.

2. ETL job failed midway

Q: How would you maintain consistency?

A:
Use checkpointing and rollback mechanisms so partial loads are not committed.
Idempotent ETL jobs allow the process to restart safely without duplicating records.

Additional DBMS Topics:

1. Audit trail of loan applications

Q: How would you support this?

A:
Enable audit logging and version control for loan applications.
Use immutable records so every change is recorded and can be traced for regulatory compliance.

2. ML model needs daily updated training data

Q: How would you manage the data pipeline?

A:
Use incremental ETL pipelines and daily data snapshots to provide fresh loan default data for model training efficiently.

3. Duplicate credit card records

Q: How would you detect and resolve them?

A:
Apply data quality checks and fuzzy matching based on name, phone number, and date of birth during data transformation to identify and remove duplicate records.

4. Real-time analytics dashboards

Q: What DBMS architecture would you propose?

A:
Use a Hybrid Transactional and Analytical Processing (HTAP) architecture with:

OLTP database for transactions, OLAP system for analytics

Materialized views for fast queries

Caching (Redis) for real-time dashboard performance

This supports live dashboards for transactions, credit scores, and investments.

SQL AND EXCEL WORK APPROACH

Section 1: Project Overview & Objectives

Project Overview

The *Financial Risk and Performance Analysis in Consumer Lending* project focuses on analyzing loan data to understand risk patterns, borrower behavior, and overall lending performance.

In the consumer lending industry, financial institutions face significant risk due to loan defaults, changing interest rates, and varying borrower income levels. This project aims to evaluate these risk factors using structured data analysis techniques.

Through SQL-based ETL processes and Excel dashboard visualization, we analyzed loan amounts, interest rates, loan purposes, regional distributions, income categories, and default trends. The goal is to generate actionable insights that help financial institutions minimize risk and optimize lending strategies.

The project combines:

- Data Cleaning and Transformation (ETL using SQL)
- Exploratory Data Analysis (Excel)
- Dashboard Creation
- Business Insight Generation

Project Objectives

The key objectives of this project are:

1. **To analyze loan distribution patterns** across different regions, purposes, and income categories.
2. **To evaluate default risk (Bad Loans vs Good Loans)** and understand factors influencing loan performance.
3. **To study interest rate trends over time** and across different loan terms.
4. **To assess the relationship between annual income and loan amount** to understand borrower risk profiles.
5. **To identify high-risk loan segments** and provide strategic recommendations to reduce financial losses.
6. **To develop an interactive Excel dashboard** that allows dynamic filtering and real-time insight exploration.

Section 2: ETL Phase (SQL)

The complete ETL process including:

- **Table creation**
- **Data loading**
- **Data validation**
- **Data cleaning**
- **Feature engineering**
- **Creation of analytical columns**
- **Default rate calculation**

has been implemented in PostgreSQL.

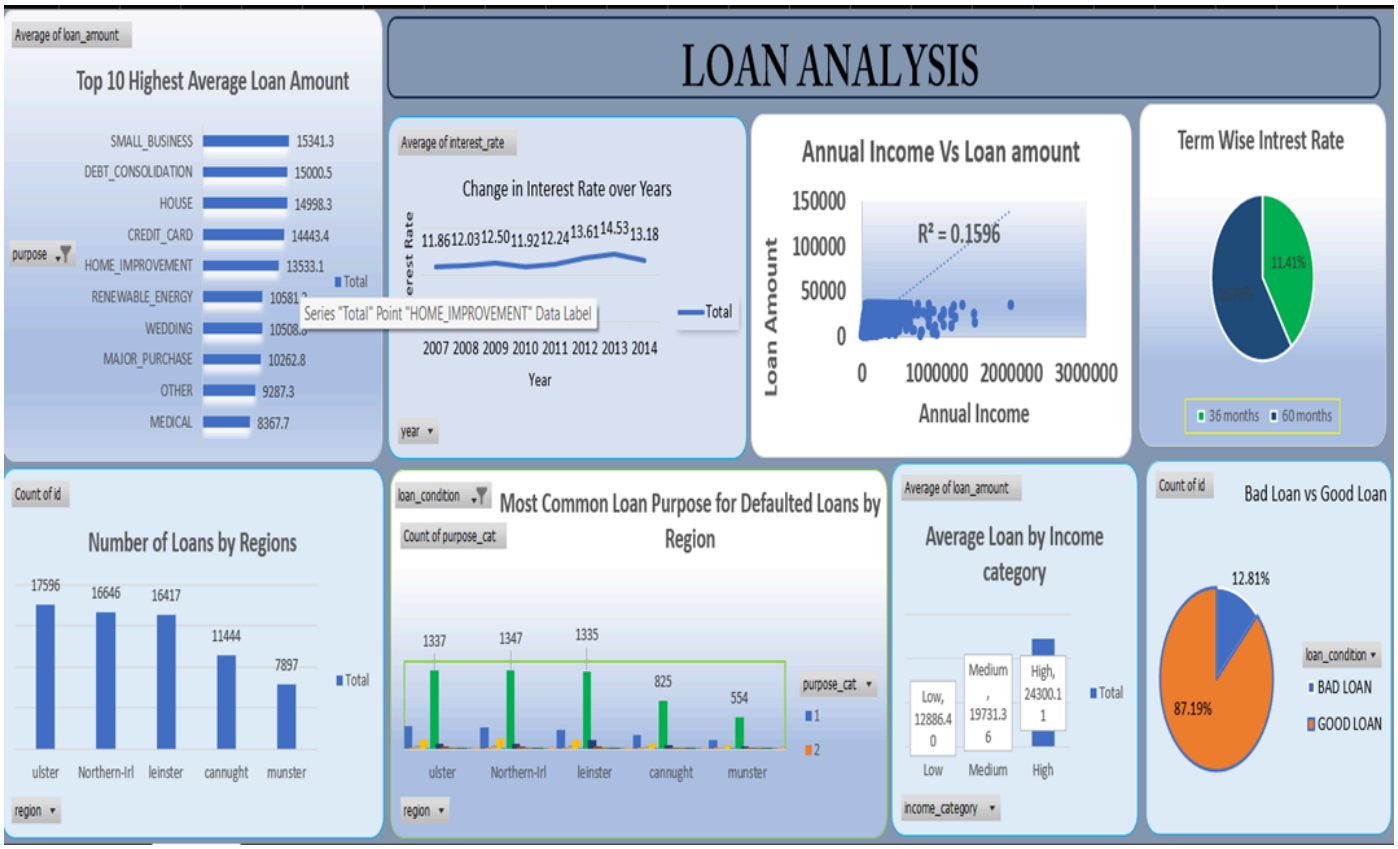
🔗 **Colab / PostgreSQL Environment Link:**

https://drive.google.com/file/d/1fYT-QFYI0xfmWrPWXz_exwig5RXJ9Ddg/view?usp=drive_link

Section 3: Excel Analysis

Excel File Link (All Questions Completed):

loan_excel_data_new_.xlsx



Section 4: Insights & Observations

Top Findings:

- Majority of loans fall under good loan category with a smaller portion classified as bad loans.
- Certain regions show higher default concentration.
- Interest rates show variation across loan terms.
- Income level significantly impacts loan amount and risk profile.
- Some loan purposes have higher average loan amounts.

Recommended Business Actions:

1 Strengthen Risk Assessment for High-Default Regions

The regional analysis identified specific regions where bad loan frequency is significantly higher. Financial institutions should apply stricter credit screening, enhanced verification processes, and risk scoring adjustments in these regions to reduce default probability and improve portfolio stability.

2 Implement Income-Based Credit Evaluation Models

SQL-based exploratory analysis revealed patterns between borrower characteristics and loan performance. Incorporating income-based risk thresholds and debt-to-income validation models can help ensure repayment capacity is accurately assessed before loan approval.

3 Monitor High-Risk Loan Purposes Closely

The Excel frequency analysis identified certain loan purposes that dominate bad loan cases in specific regions. These high-risk categories should be monitored through tighter approval criteria, periodic performance reviews, and possibly revised lending caps to control exposure.

4 Optimize Interest Rate Strategy for Different Loan Terms

Default patterns suggest variation in risk across different loan terms and categories. A risk-adjusted interest rate strategy can be implemented, where higher-risk loan types or regions are assigned rates aligned with their default probability, ensuring better risk-return balance.

Section 5: Reflection

Challenges Faced

Throughout the execution of this project, we encountered several practical and technical challenges that contributed significantly to our learning.

One of the primary challenges was handling a large real-world financial dataset. Ensuring data accuracy required careful validation using SQL queries to check for missing values, verify categorical consistency, and confirm structural integrity. Maintaining clean and reliable data before analysis was critical.

Another challenge was standardizing categorical variables such as *home ownership, loan condition, and purpose*. Inconsistent formatting could have led to incorrect aggregations, so data cleaning was an essential step in the ETL process.

A major learning moment occurred during dashboard development in Excel. Initially, separate pivot tables and pie charts were created independently for different analyses. When slicers were added to the dashboard, we realized that they were not connecting to all charts. This issue arose because each pivot table had its own pivot cache.

Through troubleshooting, we learned an important best practice in dashboard design:

Instead of creating multiple pivot tables separately, one master pivot table should be created and duplicated for additional visuals to ensure slicer connectivity.

Rebuilding the dashboard using duplicated pivot tables resolved the issue and significantly improved interactivity. This experience strengthened our understanding of how Excel's pivot architecture works in real-world scenarios.

Another significant challenge arose during the process of identifying the most frequent loan purpose associated with bad loans across different regions. Since loan purpose was a categorical variable, Excel's statistical functions such as MODE could not be directly applied to text values.

To perform frequency-based risk analysis, we needed to convert categorical values (loan purposes) into numerical codes without distorting the original relationships in the dataset. Initially, applying nested formulas such as IF, MODE, MATCH, and INDEX within structured table references resulted in repeated formula errors due to Excel's array formula limitations.

To overcome this challenge we breaking the problem into structured steps—first identifying the dominant category numerically and then mapping it back to meaningful business context—the analysis became both technically robust and analytically insightful.

Overall Learning Process

This project provided end-to-end exposure to the complete data analytics lifecycle, from raw data to business insights.

We gained practical experience in:

- Performing ETL operations using PostgreSQL
- Validating and cleaning financial datasets
- Engineering new analytical features such as profitability, risk flag, income-to-loan ratio, and default rate indicators
- Writing SQL queries using CTEs and aggregation techniques
- Building interactive dashboards in Excel
- Using advanced formulas and pivot-based analysis
- Applying data-driven thinking to financial risk evaluation
- Conditional aggregation in Excel
- Array-based statistical computation
- Encoding categorical variables for analytical processing
- Separating computation logic (MODE) from interpretation logic (INDEX-MATCH)

Working with a real consumer lending dataset enhanced our understanding of how financial institutions measure loan performance, assess risk, and make strategic decisions based on data.

Key Takeaways

- Importance of structured ETL before visualization
- Understanding pivot cache and slicer relationships in Excel
- Exposure to real-world financial risk metrics
- Ability to translate raw data into actionable business insights
- Improved analytical thinking and problem-solving skills

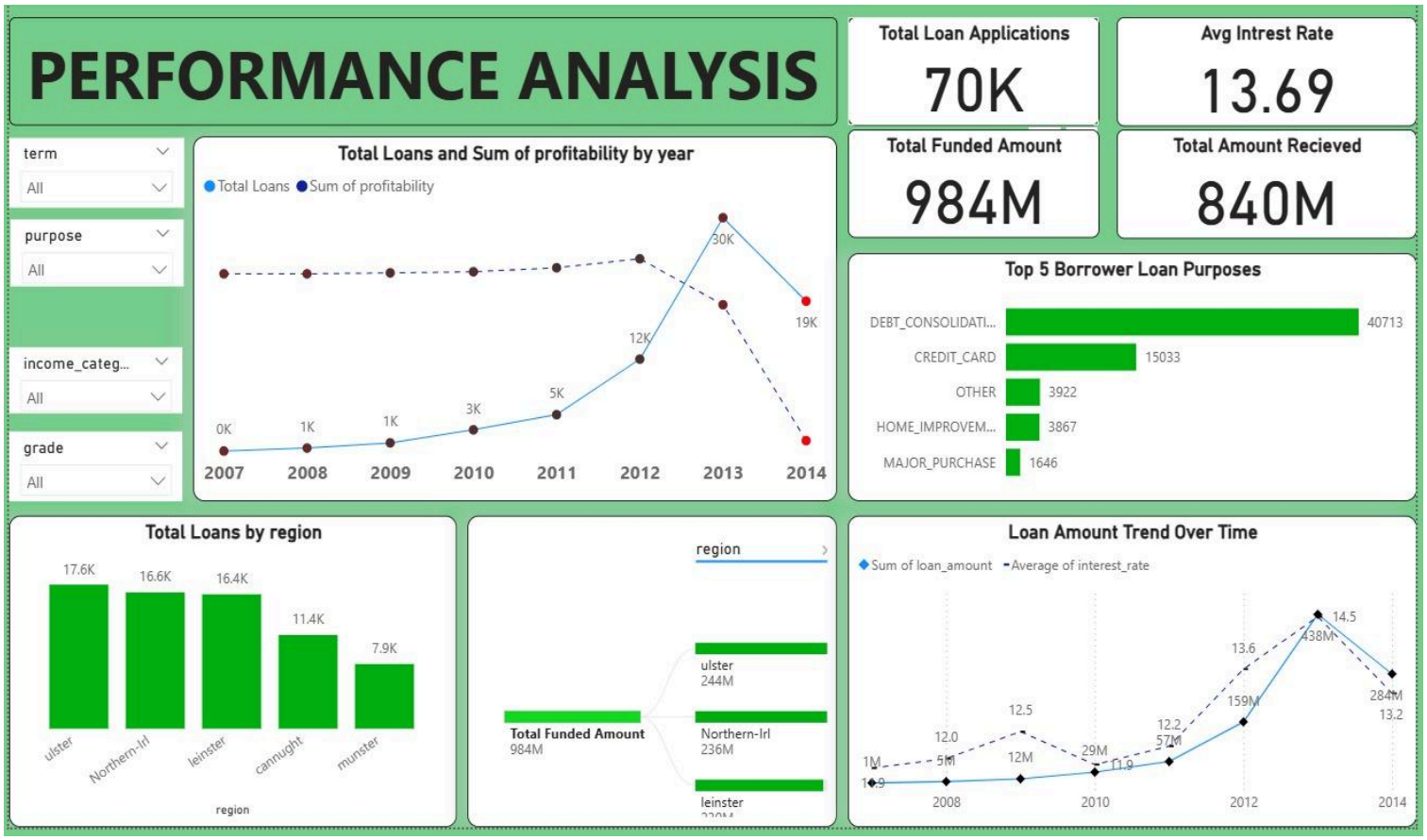
Conclusion

This project significantly enhanced our technical, analytical, and problem-solving abilities. It strengthened our confidence in managing a complete analytics workflow — from SQL-based data preparation to interactive dashboard creation and strategic recommendation development.

Overcoming practical challenges, particularly in dashboard structuring and slicer integration, was a valuable learning experience that deepened our understanding of real-world data analytics implementation.

POWER BI AND N8N AUTOMATION WORK APPROACH

Section 1: Power BI Dashboard Approach



RISK ANALYSIS

Clear Filters

region: All

year: All

term: All

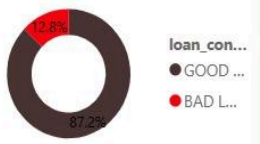
purpose: All

income_cat: All

loan_condition: All

grade: All

Good vs Defaulted Loans



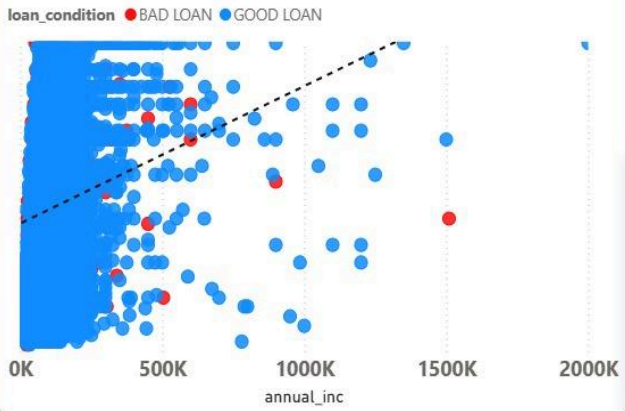
Default Rate

12.81%

Recovery Rate

5.1%

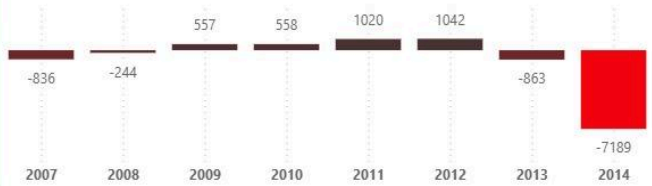
Income vs Loan Amount – Default Risk Analysis



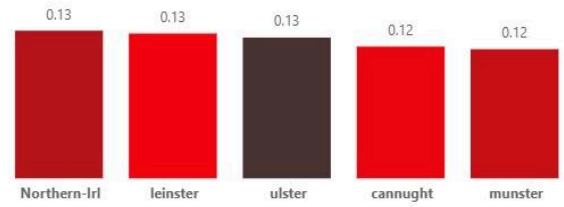
Risk Matrix: Default Rates Across Grades and Employment Tenure

grade	0.50	1.00	2.00	3.00	4.00	5.00	6.00	6.05	7.00	8.00	9.00	10.00	Total
A	0.07	0.05	0.05	0.04	0.03	0.04	0.03	0.09	0.05	0.05	0.03	0.04	0.05
B	0.10	0.09	0.09	0.10	0.10	0.10	0.11	0.13	0.10	0.09	0.11	0.09	0.10
C	0.14	0.14	0.14	0.13	0.13	0.13	0.13	0.17	0.14	0.14	0.15	0.13	0.14
D	0.22	0.21	0.18	0.20	0.19	0.16	0.19	0.19	0.17	0.20	0.18	0.18	0.19
E	0.23	0.21	0.22	0.26	0.26	0.19	0.24	0.19	0.28	0.17	0.22	0.23	0.23
F	0.26	0.30	0.33	0.22	0.29	0.32	0.27	0.32	0.24	0.31	0.39	0.25	0.28
G	0.30	0.47	0.25	0.33	0.29	0.40	0.30	0.00	0.37	0.19	0.28	0.35	0.33
Total	**0.14**	**0.13**	**0.13**	**0.13**	**0.13**	**0.12**	**0.13**	**0.15**	**0.13**	**0.12**	**0.14**	**0.12**	**0.13**

Average Loan Profitability Across Grades



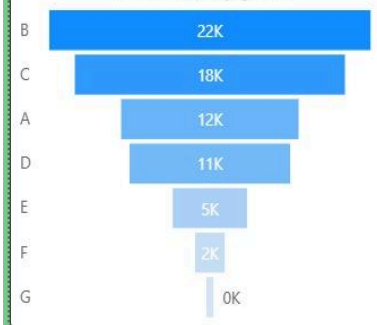
Default Rate by region



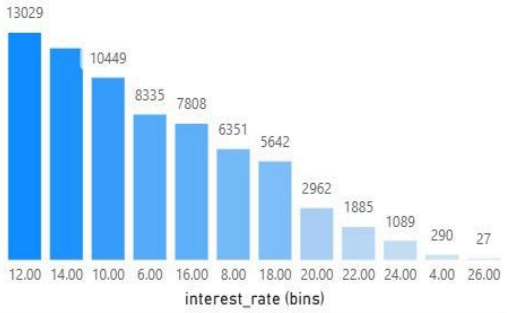
Loan Portfolio Overview

term: 36 months | 60 months

Total Loans by grade

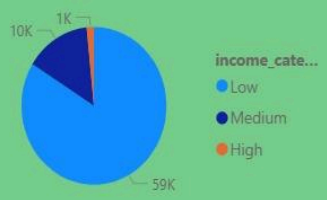


Number of Loans by Interest Rate

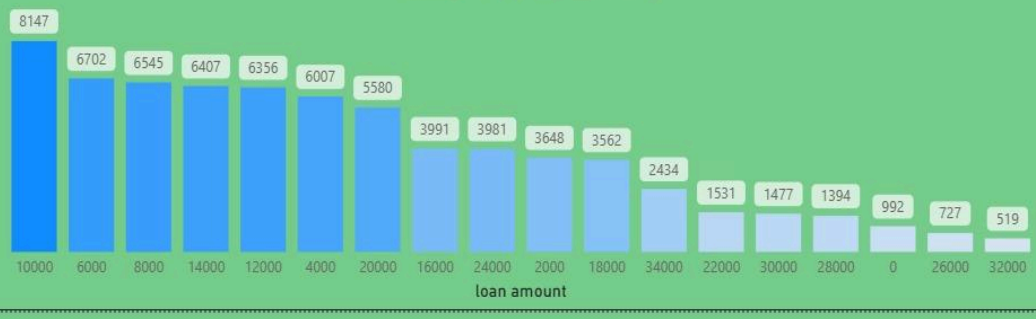


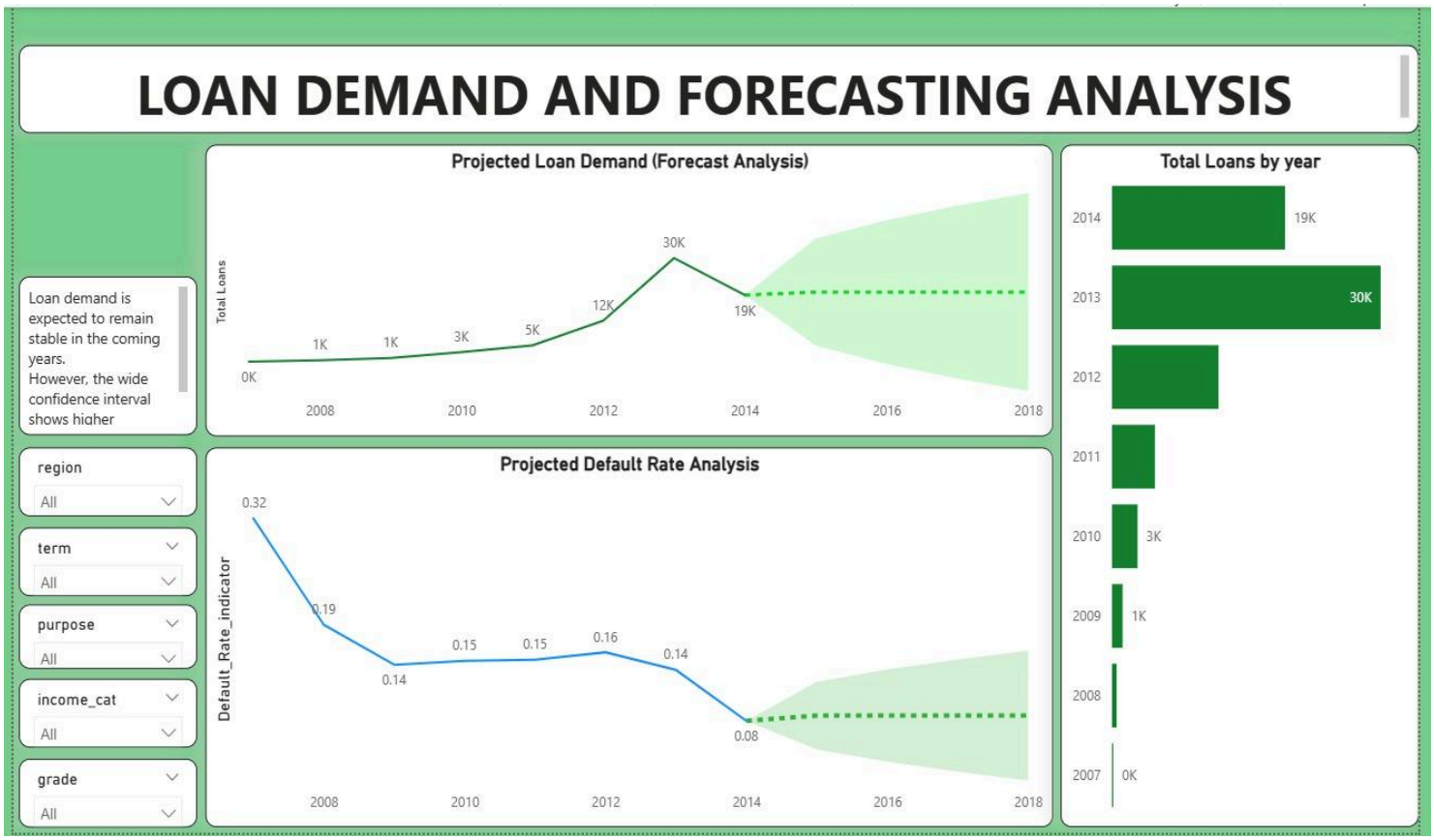
year	Sum of loan_amount	Sum of total_pymnt	Sum of profitability
2007	1.2M	1.05M	-116K
2008	4.8M	4.63M	-126K
2009	11.8M	12.49M	653K
2010	29.2M	30.81M	1,575K
2011	56.6M	61.47M	4,840K
2012	159.4M	171.68M	12,272K
2013	437.8M	412.15M	-25,619K
2014	283.6M	146.11M	-1,37,503K

Total Loans by income_category



Distribution of Loan Amounts





Section 2 : Insights

Risk Analysis – Key Insights

1. Income Level is a Major Risk Factor

- Borrowers in the **Low income category show the highest default rate** compared to Medium and High income groups.
- This indicates that lower income directly increases the probability of loan default and should be treated as a high-risk segment.

---

2. Employment Length Reduces Risk

- Customers with **shorter employment history have significantly higher default rates**.
- Borrowers with long-term employment demonstrate better repayment stability and lower credit risk.

---

3. Loan Grade Strongly Impacts Default Risk

- Loans with **poor grades (high-risk grades)** record much higher default rates than good-grade loans.
- Loan grade is one of the strongest predictors of credit risk in the dataset.

---

4. High Interest Rate Loans are Riskier

- Loans with **higher interest rates show increased default probability**.
- This suggests that risky borrowers are charged higher interest but still face difficulty in repayment.

---

5. Regional Risk Concentration

- Certain regions exhibit **consistently higher default rates**, indicating geographical concentration of credit risk.
- These regions should be closely monitored and may require stricter loan approval criteria.

---

6. Loan Purpose Influences Risk

- Loans taken for **personal expenses and debt consolidation** have higher default rates compared to business or education-related loans.
- Non-productive loan purposes contribute more to loan risk.

---

7. Debt-to-Income (DTI) Ratio and Risk

- Borrowers with **higher DTI ratios show higher default rates**, indicating over-leveraged customers are more likely to default.
- DTI is a critical financial health indicator for risk assessment.

---

8. Outliers Increase Portfolio Risk

- A small group of borrowers with **low income but high loan amounts** show extreme default behavior.
- These outliers significantly increase the overall risk exposure of the loan portfolio.

---

9. Loan Status Distribution Confirms Risk Exposure

- A noticeable portion of loans fall under **late and defaulted status**, confirming the presence of risk in the portfolio.
- Recovery rates for defaulted loans are low, increasing financial loss.

---

10. Time-Based Risk Variation

- Default rates vary across years, showing that **economic conditions influence borrower repayment capacity**.
- Periods of higher default reflect increased systemic risk in lending operations.

Business Actions

1. Stricter Screening for High-Risk Borrowers

- Apply tighter eligibility criteria for **low-income customers and borrowers with short employment history**.
- Introduce minimum income and employment stability thresholds before loan approval.

---

2. Region-Specific Risk Policies

- Implement **customized credit policies for high-default regions**.
- Increase monitoring and reduce loan exposure in regions with consistently high default rates.

---

3. Interest Rate Optimization

- Do not rely only on high interest rates to compensate for risk.
- Offer **lower interest rates to low-risk customers** and avoid over-lending to high-risk customers with excessive rates.

---

4. Loan Purpose-Based Approval Strategy

- Prioritize loans for **productive purposes (business, education)** which show lower default risk.
- Apply stricter approval rules for personal and debt consolidation loans.

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5. Debt-to-Income (DTI) Threshold Enforcement

- Set maximum acceptable **DTI ratio limits** to prevent lending to over-leveraged borrowers.
- Automatically flag applications with high DTI as high risk.

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6. Early Warning Risk Monitoring System

- Build dashboards to track **late payment patterns and rising default trends** in real time.
- Trigger alerts for customers moving from “current” to “late” status.

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7. Risk-Based Loan Amount Limits

- Cap loan amounts for **low-income or high-risk segments** to reduce potential loss.
- Allow higher loan limits only for borrowers with strong repayment history.

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8. Improve Recovery & Collection Strategy

- Focus recovery efforts on regions and segments with high default volume.
- Introduce structured repayment plans for struggling borrowers.

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9. Predictive Risk Modeling

- Use historical data to develop **predictive models for default probability** before loan approval.
- Score each customer based on risk and automate approval decisions.

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10. Customer Financial Education Programs

- Provide guidance to high-risk customers on managing debt and repayment schedules.
- This can reduce future defaults and improve customer loyalty.

Conclusion

This project analyzed loan data to understand the main factors that increase the risk of loan default. The results show that borrowers with low income, short employment history, high debt-to-income ratios, and poor loan grades are more likely to default. Higher default rates were also observed in certain regions and for loans taken for personal and debt consolidation purposes. High interest rate loans were found to be riskier, as these borrowers face more difficulty in repayment. Based on these findings, the project recommends stricter screening of high-risk borrowers, setting limits based on income and DTI, and using predictive risk models and early warning systems. Overall, this analysis helps improve loan approval decisions and supports the creation of a safer and more stable loan portfolio using data-driven strategies.